

# **Gas Storage Valuation Model: A Commodity Trading Simulation Using Time-Series Price Dynamics**

## **1. Project Overview**

This project develops a simplified gas storage valuation framework using historical natural gas price data. The model simulates trading decisions based on observed price movements to approximate the economic value of storing and trading natural gas over time.

The objective is to capture the fundamental intuition behind energy storage valuation: profitability arises from exploiting temporal price variations in commodity markets.

Unlike traditional models that rely on predefined injection and withdrawal schedules, this implementation uses a data-driven, rule-based approach derived directly from market price movements.

## **2. Financial Interpretation**

A gas storage asset provides optionality in commodity markets, allowing the holder to:

- Store gas when prices are relatively low (implicit “buy” decision)
- Release gas when prices are relatively high (implicit “sell” decision)

This optionality creates value through:

- Price volatility
- Timing of trades
- Efficient inventory management

In essence, gas storage behaves like a real option whose payoff depends on optimal timing decisions.

## **3. Problem Formulation**

The model evaluates a trading strategy based on historical natural gas prices.

## **Inputs:**

- Time-series gas prices (CSV dataset)
- Dates associated with prices
- Storage position (dynamic)
- Trading rules based on price movements

## **Objective:**

Estimate cumulative profit and loss (PnL) from a rule-based storage strategy.

Unlike classical valuation models, this approach does not assume predefined injection or withdrawal schedules. Instead, trading decisions are inferred directly from market behavior.

# **4. Trading Logic (Cashflow Mechanics)**

At each time step, the model applies the following rules:

## **Buy / Injection Proxy**

When price increases relative to the previous period:

- Gas is purchased
- Storage position increases
- Cash decreases

## **Sell / Withdrawal Proxy**

When price decreases relative to the previous period:

- Gas is sold from storage
- Storage position decreases
- Cash increases

## **Storage Tracking**

The model maintains a dynamic inventory level representing stored gas.

## **Profit Calculation**

Profit is computed as the cumulative net cashflow from all trading decisions over time.

## 5. Model Assumptions

To simplify real-world complexity, the following assumptions are made:

- No transaction frictions or bid-ask spreads
- Perfect liquidity (instant execution at observed prices)
- No storage capacity constraints
- No injection/withdrawal rate limits
- No discounting of future cashflows
- Deterministic trading rule based on price movements
- Monthly prices accurately represent market dynamics

These assumptions allow the model to focus on core price-driven behavior.

## 6. Methodology

The model operates as a sequential time-series simulation:

1. Load natural gas price data from CSV
2. Convert dates into proper time format
3. Iterate through each time step:
  - Compare current price to previous price
  - Apply trading rule (buy/sell decision)
4. Update:
  - Cash position
  - Storage position
  - Cumulative PnL
5. Store results for visualization
6. Generate performance plots

This structure mimics a simplified energy trading strategy engine.

## 7. Implementation Logic

for each time step  $i$ :

```
if price[i] > price[i-1]:
```

```
    buy gas
```

```
    increase storage
```

```
    decrease cash
```

```
elif price[i] < price[i-1]:
```

```
    sell gas
```

```
    decrease storage
```

```
    increase cash
```

```
record:
```

```
    storage level
```

```
    cumulative PnL
```

## 8. Data Description

The dataset consists of:

- Monthly natural gas prices
- Time range covering multiple years
- Structured as a time-series CSV file

Columns:

- Dates
- Prices

This dataset represents historical market movements used to simulate trading decisions.

## 9. Outputs

The model produces three key outputs:

## 1. Price Evolution

Visualizes historical gas price movements over time.

## 2. Storage Position

Tracks how inventory evolves under the trading strategy.

## 3. Cumulative Profit and Loss (PnL)

Represents total profitability of the strategy over time.

Additionally, the model outputs:

- Final PnL value
- Final storage position

## 10. Key Insight from Results

The profitability of the strategy is entirely driven by:

- Price volatility
- Frequency of reversals in price trends
- Effectiveness of the rule-based decision system

A positive PnL indicates that the strategy successfully captured short-term price movements, while a negative PnL suggests overtrading or poor timing of buy/sell signals.

Importantly, this model highlights that:

Value in commodity storage is not inherent in the asset itself, but in the ability to time market movements effectively.

## 11. Limitations of the Model

While the model captures core intuition, it simplifies many real-world complexities:

- No optimization of trading decisions
- No explicit storage constraints or capacity limits
- No injection/withdrawal costs
- No stochastic price modeling

- No forward curve or term structure
- No risk management or hedging layer

In reality, gas storage valuation is a constrained stochastic optimization problem.

## 12. Possible Extensions

This framework can be extended into a more advanced quantitative model:

### Optimization

- Linear programming for optimal buy/sell decisions
- Dynamic programming approaches

### Stochastic Modeling

- Geometric Brownian Motion (GBM)
- Monte Carlo simulation of price paths

### Advanced Pricing

- Forward curve modeling
- Real options valuation techniques

### Machine Learning

- Predict optimal trading signals
- Time-series forecasting for gas prices

## 13. Key Takeaway

This project demonstrates a simplified but realistic framework for understanding gas storage valuation through time-series trading simulation.

It highlights that commodity storage value arises from:

The ability to exploit temporal price spreads through disciplined trading decisions under uncertainty.

The model bridges:

- Financial theory (real options intuition)
- Energy trading logic
- Practical computational implementation

## 14. Interpretation of Model Visualizations

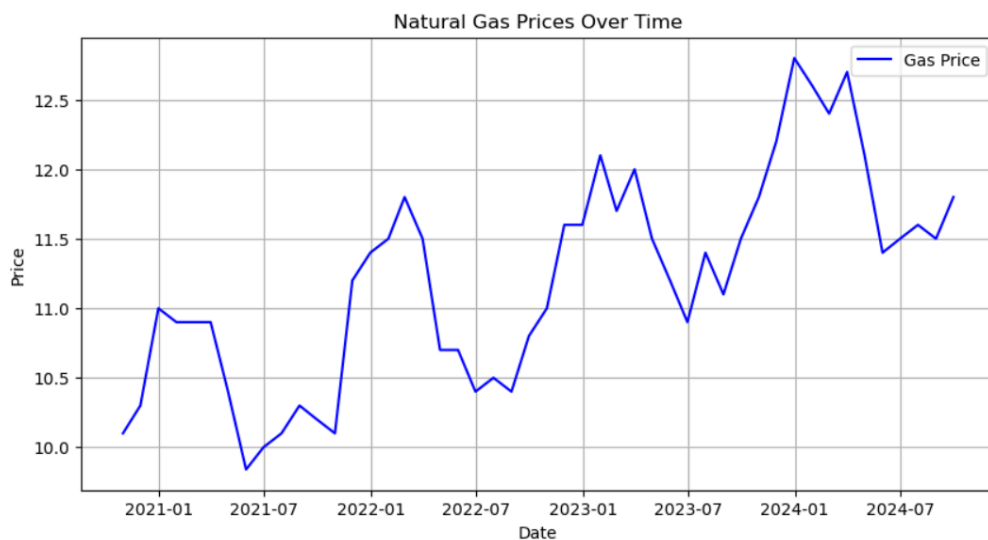
### 14.1 Gas Price Evolution

The gas price time series shows clear **fluctuations over time**, reflecting natural volatility in energy markets.

#### Interpretation:

- Periods of rising prices indicate bullish market conditions
- Periods of decline indicate oversupply or reduced demand
- Volatility is a key driver of storage value since profit depends on buying low and selling high

This confirms that the dataset contains meaningful trading opportunities.



### 14.2 Storage Level Over Time

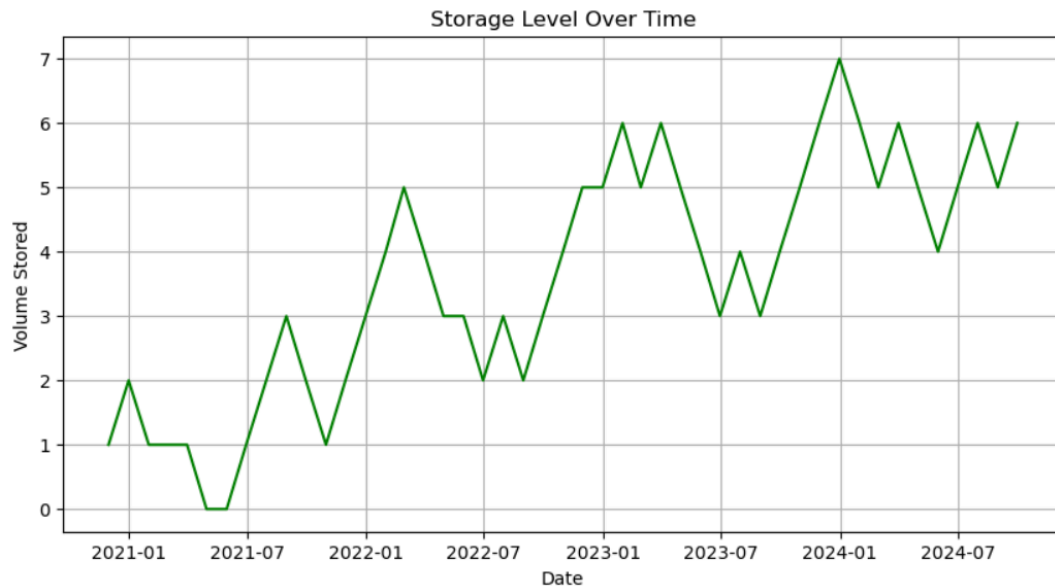
The storage level graph shows how the model dynamically accumulates and releases gas based on price movements.

#### Interpretation:

- Storage increases during upward price trends (buy decisions dominate)
- Storage decreases during downward trends (sell decisions dominate)

- The oscillation of storage reflects reactive trading behavior rather than a fixed schedule

This demonstrates that the model is actively responding to market signals.



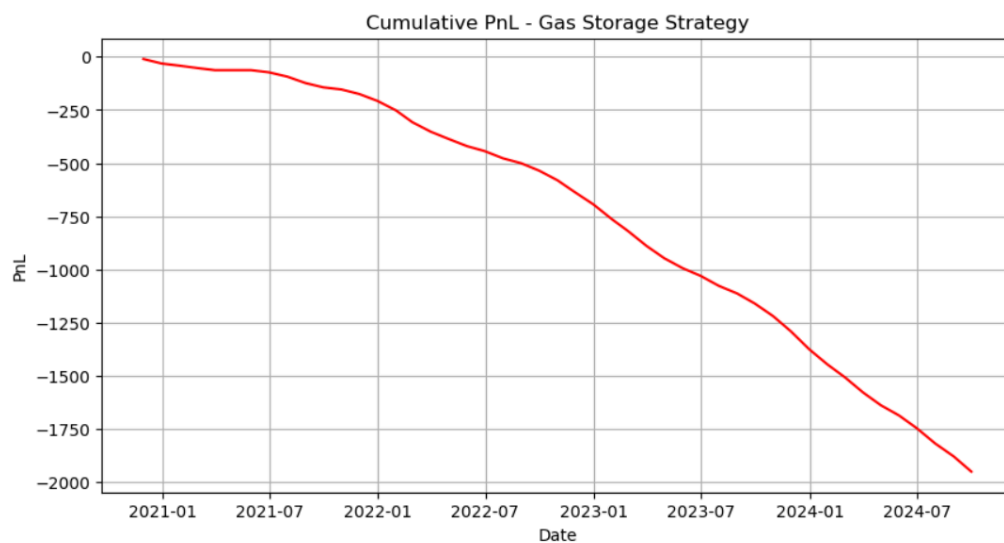
## 14.3 Cumulative PnL Curve

The cumulative profit and loss curve shows the total financial performance of the strategy over time.

### Interpretation:

- Upward sloping sections indicate profitable trading periods
- Flat or downward sections indicate inefficient timing or adverse price movements
- Overall trend determines whether the strategy successfully captures price volatility

This is the most important performance indicator of the model.





## 14.4 Insight from All Graphs

When analyzed together:

- Price movements drive trading decisions
- Storage level reflects exposure to market trends
- PnL curve captures the effectiveness of the strategy

The model behaves as a momentum-driven trading strategy where profitability depends on correctly responding to short-term price trends rather than optimizing a full storage schedule

### Final Result Interpretation

The model produced a final PnL of **-1950.50**, indicating that the rule-based trading strategy was not profitable over the observed period. This suggests that the momentum-based decision rule (buying on price increases and selling on price decreases) does not effectively capture optimal timing in the gas market.

The negative performance is primarily driven by delayed reaction to price movements and the absence of optimization in entry and exit decisions.

Additionally, the final storage level of **6 units** indicates that the strategy did not fully liquidate its position by the end of the simulation, leaving residual exposure to market risk.